Intelligent Health Systems – From Technology to Data and Knowledge
E. Andrikopoulou et al. (Eds.)

© 2025 The Authors.
This article is published online with Open Access by IOS Press and distributed under the terms of the Creative Commons Attribution Non-Commercial License 4.0 (CC BY-NC 4.0).
doi:10.3233/SHTI250472

Comparative Analysis of ChatGPT-4 for Automated Mapping of Local Medical Terminologies to SNOMED CT

Sookyung HUH[a,1]
[a] *department of medical recorder's team, Severance Hospital, Seoul, Korea*
ORCiD ID: Sookyung Huh https://orcid.org/0009-0001-8817-2004

Abstract. Standardizing medical terminology is critical for healthcare informatics, particularly for improving data interoperability and patient management systems. This study evaluated four distinct GPT-4-based approaches for mapping local medical terminologies to SNOMED CT: baseline, prompt-engineered, fine-tuned, and Retrieval-Augmented Generation (RAG). Using 1,200 diagnostic terms from a Korean hospital, we assessed the models' accuracy and error rates. The RAG model achieved the highest performance with a 96.2% valid SNOMED CT term match rate and a 57.6% overall exact match rate, surpassing the fine-tuned model (69.2% valid term match, 47.2% exact match). Error analysis showed that the RAG model also minimized structural errors to 14%, significantly lower than other models. While the fine-tuned and RAG models struggled with specificity, they showed promise for improving automated mapping and Named Entity Recognition (NER) tasks in clinical settings. This study highlights the potential of AI-human collaboration for enhancing autocoding and data standardization in healthcare. Further research is needed to refine specificity and validate these systems for clinical use.

Keywords. SNOMED CT, LLM, GPT-4

1. Introduction

Standardizing medical terminology is crucial for healthcare informatics, ensuring consistent communication across platforms and facilitating efficient patient data management [1]. SNOMED CT (Systematized Nomenclature of Medicine Clinical Terms) plays a vital role in enhancing semantic interoperability and standardization in healthcare [2-4]. The process of mapping local medical terminologies to SNOMED CT presents significant challenges. Traditionally performed manually by domain experts, this task is time-consuming, costly, and prone to inconsistencies [5]. The vast scope of medical knowledge, nuances of clinical language, and the continuous evolution of medical concepts further compound these difficulties. Recent advancements in Large Language Models (LLMs) offer potential solutions, but studies have shown poor performance in medical coding tasks, with exact match rates below 50% for ICD and CPT codes [6]. SNOMED CT presents an even greater challenge due to its extensive scope, containing about 367,700 active concepts compared to 72,000 in ICD-10-CM. Among the available LLMs, ChatGPT-4 has shown promising results in medical

[1] Corresponding Author: Sookyung Huh; E-mail: nacona1@yuhs.ac.

knowledge tasks [7]. However, its effectiveness in SNOMED CT mapping remains to be evaluated. This study aims to assess ChatGPT-4's capability in mapping local medical terminologies to SNOMED CT, comparing baseline, prompt-engineered, fine-tuned models and a Retrieval Augmented Generation (RAG) model.

2. Methods

2.1. Data Preparation and Selection

We used a dataset of 38,357 diagnostic terms from Severance Hospital, Korea. From this, 1,200 terms were randomly selected for evaluation, comprising both standard and local terms mapped to ICD-10. Terms from chapters U, V, and Z were excluded to match SNOMED CT's "Disorder" or "Clinical Finding" classes. Hospital technicians pre-mapped these terms to SNOMED CT, serving as the gold standard. For fine-tuning, an additional 2,000 terms were used for training and 400 for validation, ensuring no overlap with the evaluation set. Due to institutional policies, the dataset is unavailable publicly but can be accessed upon request via a formal agreement with the corresponding author.

2.2. Mapping Approaches

We developed and tested three approaches for mapping local terminologies to SNOMED CT using GPT-4 (September 2024 version), chosen for its enhanced language understanding. For all models, we set the temperature to 0.2 and limited output to 50 tokens, ensuring consistent responses without truncation.

1. Baseline GPT-4 Model: We employed a simple prompt structure requesting the model to return the closest SNOMED CT term from the 'Disorder' and 'Clinical Finding' categories for a given medical diagnosis.
2. Prompt-Engineered Approach: We iteratively developed an enhanced prompt incorporating examples and specific instructions, following the principles outlined in OpenAI's prompt engineering guide [8]. This approach was designed to improve mapping accuracy by providing context and constraints. The prompt included 10 carefully selected example mappings and detailed guidelines for term selection.
3. Fine-tuned Model: We fine-tuned the GPT-4 base model (gpt-4-0613) using a dataset of 2,000 training examples and 400 validation examples. These examples were carefully curated to represent a diverse range of medical terminologies and their corresponding SNOMED CT mappings. The fine-tuning process employed the following hyperparameters: 6 epochs, batch size of 8, learning rate multiplier of 2, and a random seed of 42. This resulted in a model with a training loss of 0.0053 and a validation loss of 0.5560. The fine-tuned model was then applied using the prompt-engineered approach.
4. Retrieval-Augmented Generation (RAG) Model: We implemented a Retrieval-Augmented Generation (RAG) Model that combines retrieval and GPT-4's generation capabilities. The model retrieves the top 5 most similar SNOMED CT terms using cosine similarity and TF-IDF, and GPT-4 selects the most relevant term or provides a valid approximation if none fit, enhancing terminology mapping accuracy.

2.3. Evaluation Metric

We evaluated our mapping approaches using the following metrics:

1. Mapping Accuracy:
   - Valid SNOMED CT Term Match (%): Percentage of terms mapped to valid SNOMED CT terms.
   - Overall Exact Match (%): Percentage of terms exactly matching human-mapped terms.
   - Exact Match Among Valid Matches (%): Exact matches within valid mappings.
2. Semantic Similarity:
   - Cosine Similarity: Measures semantic similarity between predicted and reference terms.
   - METEOR Score: Evaluates precision and recall of term mapping.
   - BERT Score: Assesses contextual similarity.
3. Error Analysis: (on 200 manually reviewed mappings):
   - Semantic Errors: Incorrect or ambiguous concept selection.
   - Specificity Errors: Over- or under-specificity in mapping.
   - Structural Errors: Incorrect class assignments or non-existent codes.
   - Terminology Management Errors: Synonym matches and variations.

3. Results

3.1. Overall Performance Metrics

Table 1 illustrates the overall performance metrics of each model. The RAG model showed superior performance, achieving a 96.2% valid SNOMED CT term match rate, compared to 69.2% for the fine-tuned model, 51.7% for the prompt-engineered model, and 39.8% for the baseline model. The RAG model's success is due to its retrieval-augmented method, where relevant terms are first retrieved via cosine similarity, and GPT-4 refines the selection, leading to a higher match rate. The RAG model also outperformed others in overall exact match rate at 57.6%. Despite lower cosine similarity (0.08) compared to the fine-tuned model (0.4), the RAG model excelled in overall valid and exact matches. It also performed competitively in METEOR and BERT scores (0.56 and 0.92, respectively), demonstrating strong contextual alignment.

3.2. Manually Reviewed Subset Analysis

To gain deeper insights into the models' performance and error patterns, we conducted a detailed analysis on a manually reviewed subset of 200 mappings (Table 2). In the 200 manually reviewed mappings, the RAG model achieved a 94% valid term match rate, exceeding the fine-tuned model (70.5%), prompt-engineered model (46.5%), and baseline model (43%). This was largely attributed to the RAG model's ability to first retrieve valid terms, then refine them with GPT-4. The RAG model also led in overall exact match (57.2%). In terms of errors, the RAG model had the lowest rate of incorrect codes (43%), while slightly more semantic errors (4.7%) compared to the fine-tuned model (3.1%). Specificity errors were also high (30.2%), mostly due to under-specificity, where general terms were selected over precise ones. Structural errors were significantly reduced to 14%, compared to 59.2% for the fine-tuned model, and terminology management errors, mostly due to synonym matches, were highest for the RAG model (51.2%). While synonym matches are not true errors, they can affect exact match precision.

Table 1. Overall Performance Metrics

Metric	Baseline Model	Prompt Engineered Model	Fine-tuned Model	RAG Model
Valid SNOMED CT Term Match (%)	39.8	51.7	69.2	96.2
Overall Exact Match (%)	22.5	31.8	47.2	57.6
Exact Match Among Valid Matches (%)	56.5	61.6	68.2	59.9
Cosine Similarity	0.15	0.24	0.4	0.08
METEOR Score	0.49	0.53	0.61	0.56
BERT score	0.91	0.92	0.94	0.92

Notes: Valid SNOMED CT Term Match (%): The percentage of local terms that are successfully mapped to a valid SNOMED CT term.
Overall Exact Match (%): Percentage of local terms exactly matching the manual human-mapped SNOMED CT term.
Exact Match Among Valid Matches (%): Percentage of exact matches considering only terms that had a valid SNOMED CT mapping.
Cosine Similarity: Measures the semantic similarity between the predicted term and the reference term.
METEOR Score: Evaluates the precision and recall of the term mapping process.
BERT Score: Measures the contextual similarity between the local term and the predicted SNOMED CT term.

Table 2. Manually reviewed code subset Analysis of Mappings (N=200)

Metric	Baseline Model (%)	Prompt Engineered Model (%)	Fine-tuned Model (%)	RAG Model (%)
Valid SNOMED CT Term Match (%)	**43**	**46.5**	**70.5**	**94**
Overall Exact Match (%)	**26.5**	**31**	**51**	**57.2**
Exact Match Among Valid Matches (%)	**61.6**	**66.7**	**72.3**	**61.2**
Incorrect Codes (%)	**73.5**	**69**	**49**	**43**
a) Semantic Errors	**0.7**	**2.9**	**3.1**	**4.7**
Incorrect Concept Selection	0	0	2	3.5
Ambiguous Concept Selection	0.7	1.4	1	0
Contextual Error	0.7	1.4	0	1.2
b) Specificity Errors	**12.2**	**13**	**30.6**	**30.2**
Over-Specificity (Over-Mapping)	0	0	6.1	10.5
Under-Specificity (Under-Mapping)	10.2	11.6	24.5	19.8
Partial Match	2	1.4	0	0
c) Structural Errors	**77.6**	**76.8**	**59.2**	**14**
Incorrect Concept Class	2	1.4	1	0
Non-Existent/Fabricated/Outdated Code	76.2	75.4	58.2	14
d) Terminology Management Errors	**8.2**	**7.2**	**7.1**	**51.2**
Synonym Match	8.2	7.2	7.1	51.2

Notes: Valid SNOMED CT Term Match (%): The percentage of local terms that are successfully mapped to a valid SNOMED CT term.
Overall Exact Match (%): Percentage of local terms exactly matching the manual human-mapped SNOMED CT term.
Exact Match Among Valid Matches (%): Percentage of exact matches considering only terms that had a valid SNOMED CT mapping.
Incorrect Codes (%): The percentage of terms that were incorrectly mapped.
a) Semantic Errors: Errors related to incorrect or ambiguous interpretations of concepts.
b) Specificity Errors: Errors where the model either over-specifies, under-specifies, or partially matches the concept.
c) Structural Errors: Errors involving incorrect concept classes or fabricated/outdated codes.
d) Terminology Management Errors: Errors due to synonyms or terminology variations.

4. Discussion

This study compared various GPT-4-based approaches for mapping local medical terminologies to SNOMED CT. The RAG model demonstrated the best performance, with a 96.2% valid SNOMED CT term match rate and a 57.6% overall exact match rate. The success of the RAG model is attributed to its hybrid approach, which integrates dictionary-based retrieval with GPT-4's generative refinement. While both the RAG and fine-tuned models struggled with synonym matches, the RAG model reduced structural errors significantly (14%) and had the lowest incidence of incorrect codes (43%), outperforming the other models across key metrics. Our findings align with recent work on LLMs in healthcare. Kather et al. (2024) [9] highlight large language models as a universal interface for tasks like terminology mapping, while Soroush et al. (2024) [6] underscore AI's potential in medical coding and standardization.

5. Conclusions

This study demonstrates that integrating retrieval-augmented generation (RAG) with GPT-4 improves SNOMED CT terminology mapping. The RAG model achieved the highest term match rates and greatly reduced structural errors, making it a valuable tool for healthcare terminology standardization. Both the RAG and fine-tuned models show strong potential for automated mapping and Named Entity Recognition (NER) tasks. The study emphasizes the importance of combining AI and human expertise for accurate medical coding. Further research should focus on improving specificity and developing robust validation processes for clinical use.

References

[1] Wasserman H, Wang J. An Applied Evaluation of SNOMED CT as a Clinical Vocabulary for the Computerized Diagnosis and Problem List. AMIA Annu Symp Proc. 2003;2003:699-703.

[2] Lee D, de Keizer N, Lau F, Cornet R. Literature review of SNOMED CT use. J Am Med Inform Assoc. 2014 Feb;21(e1):e11-9. doi: 10.1136/amiajnl-2013-001636.

[3] Alahmar A, AlMousa M, Benlamri R. Automated clinical pathway standardization using SNOMED CT-based semantic relatedness. J Am Med Inform Assoc. 2022 May 11;29(6):1157-1167. doi: 10.1093/jamia/ocac037.

[4] Vuokko R, Vakkuri A, Palojoki S. Systematized Nomenclature of Medicine–Clinical Terminology (SNOMED CT) Clinical Use Cases in the Context of Electronic Health Record Systems: Systematic Literature Review. JMIR Med Inform. 2023 Feb 6;11:e43750. doi: 10.2196/43750

[5] Australian Digital Health Agency. SNOMED CT-AU Mapping Guidelines v2.0. National Clinical Terminology Service; 2014. Available from: https://developer.digitalhealth.gov.au/resources/snomed-ct-au-mapping-guidelines-v2-0

[6] Soroush A, Glicksberg BS, Zimlichman E, Barash Y, Freeman R, Charney AW, Nadkarni GN, Klang E. Large Language Models Are Poor Medical Coders — Benchmarking of Medical Code Querying. NEJM AI. 2024 Apr 19;1(5). doi: 10.1056/AIdbp2300040

[7] Kung TH, Cheatham M, Medenilla A, Sillos C, De Leon L, Elepaño C, et al. Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. PLOS Digit Health. 2023 Feb 9;2(2):e0000198. doi: 10.1371/journal.pdig.0000198.

[8] OpenAI. Prompt engineering. OpenAI Platform Documentation. https://platform.openai.com/docs/guides/prompt-engineering.

[9] Kather JN, Ferber D, Wiest IC, Gilbert S, Truhn D. Large language models could make natural language again the universal interface of healthcare. Nat Med. 2024 Aug 23;30:1487. doi: 10.1038/s41591-024-03199-w.